

Research Proposal Writing

Chapter 4 Research Fundamental

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Outline

1 Research Proposal Writing

2 Research Proposal

Research Proposal Writing

Research Proposal Writing

- Science aims to find patterns, trends, and structures in the experiment
- Good scientific writing aims to bring forward in the text
 - Structure
 - Organization of knowledge

What's in the Research?

- Addresses a problem or series of problems
- Describes what was known about the problem(s)
 - What did you do to solve the problems?

Research Proposal

- A plan of action
- A prerequisite for research investigation

Subject or Title	Set out the broad topic you would like to research
↓	
Aims and Objectives	What research would set out to achieve
↓	
Methodology	How you would go about researching it
↓	
Expected outcomes	What the results might be in relation to knowledge and understanding in the subject
↓	
Outline plan	How you would undertake it within the time available

Purpose of Research Proposal

- Research proposal is intended to **convince others** that you have
 - a **worthwhile research project**
 - the **competence** and the **work-plan to complete** it
 - **Problem is significant and worthy of study**
 - **Approach is novel and likely to give results**

- Title reflects the content of the research
- Be Specific and Accurate
 - Avoid phrases such as “A study of ...”, “An investigation of ...”

"An early identification of Arrhythmia by machine learning using wearable Electrocardiogram monitoring system"

Introduction of Research Proposal

- **General introduction:** what is the research about
- **Summarize:** research questions/problems
- **Indicate:** why is this a worthwhile problem
- Can **general readers** easily **understand the introduction?**
- Did you **explain terms** to **avoid readers' confusion?**
- Introduction may include a few sections

Introduction of Research Proposal

Introduction should include four basic components

- 1 Meaningful presentation of the study
- 2 Background of the study
- 3 Statement of the main issue of the study

Research background

- Addresses why the proposed work is important in the field: **answers the question**
- State-of-the-art of the relevant work field
 - **identify a gap in knowledge or activities** that must be filled to move the field forward
 - **to show why the previous work needs to be continued**
- Sufficient details should be given to make clear what the research problem is and exactly what has been accomplished
 - **to give evidence of your own competence in the field**
 - make summary of literature reviewed including knowledge gap

Research background

- Give the background to let the readers know and understand your topic & the knowledge of this area
- The state of the art; related work, existing research
- Organize these by logical and reasonable ideas, **not simply by time/author, etc**
- Is the **background** **clear enough?**
- Is the **literature review** **well organized and enough?**

Key questions to ask yourself: Have I

- identified the specific research problem I wish to investigate?
- indicated what I intend to do about this problem?
- put forth an argument as to why this problem is worthy of investigation?
- made my assumption explicit?

Research Questions/Problem Statements

- **Concise statement:** problems you want to address
- **Justified problem:** unanswered/not well solved based on the previous literature review
- **Thoroughly discuss:** why is it worthwhile to solve this problem?
- Will readers be convinced to **agree** with your **view on the problems, about the originality, novelty and applicability?**
- Did you refer to some **literature references** to **convince your points?**

Objectives of the Study

- Objectives should be clearly stated and specific in nature
- Avoid words “to determine”, “to find out”
- Objectives should be attainable, measurable, achievable and testable

Scope of the Study (Significant)

Significance of the study contains three paragraphs based on three questions:

- Why is this study important?
- How is this study important?
- For whom is this study important?

Methodology: Design

Proposal needs to answer following questions:

- ① What kind of data or sources will you use?
 - Clearly indicate the methods of data collection **general plan for data collection**
 - **How many?** and **From where?**
- ② Which theoretical and methodological techniques will you use to interpret and analyse these data?
 - **why you chose particular algorithm to solve the problem**
 - How does it differ from the state-of-the-art?
- Show proposed **block diagram** to show way of solving the problem
 - explanation of each block
- It is **important** that you *explain* the design of your proposal in a **clear and logical way**
- Examiner should be able to **clearly see** what you will do and how will you do it
 - How the **combination of data and methods** will allow you to address your research problem

Expected Outcome

- Conclude your research proposal by stating your expected outcomes
- What **arguments** and **conclusions** do you **expect to reach**?
- Examiner will understand that **expected outcomes based on the extent of research at the time of writing**
 - Output will **almost certainly change** in the light of further research
- This will allow an examiner to further assess the significance and validity of your research proposal

- Timeline demonstrates to the examiner that your research can be completed within the period
- Timeline should consist of a series of goals that you will need to meet in order to complete all aspects of your thesis
 - from initial research to the final editing
 - with an expected date of completion for each step